YI-LEI LI

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Summary

Software engineer with an **M.S. in Computer Science** (University of Delaware, May 2026) who ships tested, production-style systems: a deployed **React + TypeScript** app on **Node.js / PostgreSQL** with its own offline sync engine, a **Java 21 / Spring Boot 3** banking service with atomic transfers, and a **TypeScript** pipeline pulling 22,000+ records per run. Prior backend internship at Cathay Financial Holdings.

Technical Skills

Languages:	Java, TypeScript/JavaScript (Node.js), Python, SQL, HTML5/CSS3
Frontend:	React 19, React Router, Vite, Tailwind CSS, PWA (offline, service worker), IndexedDB (Dexie)
Backend:	Spring Boot 3, Node.js (Hono), JPA/Hibernate, Drizzle ORM, REST APIs, pessimistic locking, Actuator
Databases:	PostgreSQL (Neon, Row-Level Security), H2, DuckDB; window functions, CTEs
Testing & DevOps:	JUnit 5, MockMvc, Vitest, Playwright, ESLint, Maven, CI/CD (GitHub Actions, Render), Git/GitHub
ML & AI:	LLM integration (Anthropic Claude API), scikit-learn, XGBoost, SHAP, AI-assisted development

Projects

LeetCode Coach — Offline-First Interview-Prep Web App — React, TypeScript, Hono, PostgreSQL [Live] [GitHub]

- Full-stack **PWA** that my friends and I use daily for interview prep: a **React 19 + TypeScript** front end on IndexedDB over a **Node.js (Hono)** API on **PostgreSQL** (Drizzle ORM) on Render + Neon; every change ships through a pull request and CI.
- Built an **offline-first sync engine**: every local write records an outbox entry in the same transaction; the server applies **last-write-wins** upserts with tombstones and assigns versions from a Postgres sequence under a **per-user advisory lock**, so a sync cursor never skips a change.
- Implemented **FSRS-6 spaced repetition** (Anki's scheduler) from its formulas, verified against the official `ts-fsrs` on hundreds of random histories; a **sprint planner** pulls problems likely forgotten by interview day into review.
- Auth with **scrypt**-hashed passwords, hashed session tokens, CSRF checks, and rate limits; **Claude API** feedback limited to allow-listed accounts with a daily quota to cap cost. **353 unit/API** and **108 end-to-end** tests gate every deploy.

DSA Whiteboard — Open-Source React Package — React, TypeScript, Cloudflare Workers [Live] [GitHub]

- Data-structure whiteboard built from scratch (no drawing library) and **extracted from LeetCode Coach into its own package**: 20+ elements (arrays, trees, graphs, heaps, DP tables, . . .), step playback, and line-by-line code tracing; a shared **zod schema** validates boards on client and server.
- **Live co-editing** for up to 8 people: a **Cloudflare Worker with one Durable Object per room** relays per-element changes over WebSockets (last write wins); offline edits replay on reconnect. **183 unit tests** plus Playwright.

Banking Transactions API — Java 21, Spring Boot 3, PostgreSQL [GitHub]

- RESTful core-banking service with **atomic transfers** — both legs commit or neither does (`@Transactional` rollback); **ordered pessimistic row locks** (`SELECT ... FOR UPDATE`, ascending-id order) prevent double-spend and deadlock, and every movement writes an **immutable audit-ledger** row.
- **21 tests in GitHub Actions CI**, including a **concurrency race on real PostgreSQL** (Testcontainers): 100 opposite transfers at once, with no deadlock, no money created, and no negative balance.

Job-Opening Monitor — Node.js, TypeScript, Supabase [GitHub]

- Scheduled **automation pipeline** pulling **22,000+ records per run** from 115 employer and job-board sources through **11 platform adapters** (Workday, Greenhouse, Lever, Ashby, iCIMS, Oracle Cloud, . . .) behind one normalization layer, with bounded-concurrency fetches and per-source failure isolation.
- **Idempotent** re-runs via a Postgres state table; runs unattended on a cron with Telegram + email alerts. Its Claude API screening stage **fails closed**: after exhausted API credit let 82 unscreened results out, errors now hold them back. **350 Vitest tests**.

Education

University of Delaware — Newark, DE
Master of Science in Computer Science — GPA 3.66/4.0 — *Aug 2024 – May 2026*

National Kaohsiung University of Science and Technology — Kaohsiung, Taiwan
Bachelor of Science in Electrical Engineering — *2020*

Experience

Independent Engineering Projects — building and operating the projects above, Newark, DE — *May 2026 – Present*

Cathay Financial Holdings Co., Ltd. — Back-End Software Engineering Intern, Taipei — *Jul 2023 – Nov 2023*

- Delivered a CRUD service (**Java + SQL**) on an **Agile** team; traced the transaction flow end to end (balance checks, per-account concurrency control, **ACID**).

Graduate Research Assistant, Chang Gung University — **NeRF** 3D scene reconstruction — *2021 – 2022*

Undergraduate Research Assistant, NKUST IoT Lab — live sensor dashboard (**REST/WebSocket**) — *2018 – 2020*